

Op-Amps & Comparators: LM741, LM358, LM324, TL071, LM393, LM386, NE5532

Document: EDU-ANALOG-01 | Pinouts, Configurations & Application Circuits Reference

1. LM741 — Classic Single General-Purpose Op-Amp (Texas Instruments / STMicro)

Package: 8-pin DIP. Supply Voltage: +/-5V to +/-18V (dual supply) or 10V to 36V (single supply). Input Offset Voltage: 1mV typical. Input Bias Current: 80nA. Gain-Bandwidth Product: 1 MHz. Slew Rate: 0.5 V/us. Output Short-Circuit Protected. The 741 is the foundational op-amp taught in all analog electronics courses.

Pinout (DIP-8): Pin 1 = Offset Null, Pin 2 = Inverting Input (IN-), Pin 3 = Non-Inverting Input (IN+), Pin 4 = V- (negative supply / GND in single supply), Pin 5 = Offset Null, Pin 6 = Output (VOUT), Pin 7 = V+ (positive supply), Pin 8 = No Connect. Pins 1 and 5 allow external null potentiometer to zero the offset voltage using a 10k pot.

Inverting Amplifier: $\text{Gain} = -(R_f/R_{in})$. Connect IN+ to GND through $R_{comp} = R_f || R_{in}$. $V_{OUT} = -(R_f/R_{in}) * V_{IN}$.
Non-Inverting Amplifier: $\text{Gain} = 1 + (R_f/R_{in})$. Connect VIN directly to IN+. $V_{OUT} = (1 + R_f/R_{in}) * V_{IN}$.
Voltage Follower (Buffer): $\text{Gain} = 1$. Connect VOUT directly to IN-. $V_{OUT} = V_{IN}$.

2. LM358 — Dual Low-Power Op-Amp (Texas Instruments)

Package: 8-pin DIP. Contains two independent op-amp circuits in one chip. Supply: 3V to 32V single supply OR +/-1.5V to +/-16V dual supply. Input Common-Mode Range extends to GND (rail-to-rail input). Output can swing to within 1.5V of V+. Quiescent Current: 0.5mA per amplifier. Gain-Bandwidth: 1 MHz.

Pinout (DIP-8): Pin 1 = OUT-A, Pin 2 = IN-A, Pin 3 = IN+A, Pin 4 = GND (V-), Pin 5 = IN+B, Pin 6 = IN-B, Pin 7 = OUT-B, Pin 8 = VCC (V+). The LM358 is ideal for single-supply Arduino projects running at 5V because the input common-mode range includes 0V.

Comparator Application: Connect threshold voltage to IN+ via voltage divider. Connect signal to IN-. When $\text{signal} < \text{threshold}$, VOUT goes HIGH. When $\text{signal} > \text{threshold}$, VOUT goes LOW. Add 100k feedback resistor from VOUT to IN+ for hysteresis (Schmitt trigger behavior) to prevent oscillation at the threshold.

3. LM324 — Quad General-Purpose Op-Amp (Texas Instruments)

Package: 14-pin DIP. Contains FOUR independent op-amp circuits in a single package. Supply: 3V to 32V single supply or +/-1.5V to +/-16V dual supply. Pinout groups: Amplifier A (Pins 1,2,3), Amplifier B (Pins 5,6,7), Amplifier C (Pins 8,9,10), Amplifier D (Pins 12,13,14). Shared power: Pin 4 = GND, Pin 11 = VCC.

The LM324 is widely used in audio mixers, summing amplifiers, active filter designs, and multi-channel sensor signal conditioning. Each amplifier is identical to the LM358 single section. Advantages over using four separate ICs: reduced board space, single power connection, lower cost.

4. TL071 / TL081 — JFET-Input Single Op-Amp (Texas Instruments)

Package: 8-pin DIP. JFET input stage provides extremely high input impedance (10^{12} ohms) and very low input bias current (30pA typical). Supply: +/-5V to +/-18V. Slew Rate: 13 V/us (much faster than LM741). Gain-Bandwidth: 4 MHz. Input Offset Voltage: 3mV. The high slew rate makes TL071/081 essential for audio amplifiers and oscillators where waveform fidelity matters.

TL072 is the dual version, TL074 is the quad version. Pinout is identical to LM741 and LM358 respectively. The JFET input eliminates loading effects on high-impedance sources like piezo sensors, guitar pickups, and pH probes.

5. LM393 / LM339 — Voltage Comparators (Texas Instruments)

LM393: Dual voltage comparator. LM339: Quad voltage comparator. Package: 8-pin DIP (LM393), 14-pin DIP (LM339). Supply: 2V to 36V single or +/-1V to +/-18V dual. The critical difference from op-amps: comparator outputs are OPEN COLLECTOR. The output pin is the collector of an NPN transistor; it can only pull LOW or float HIGH. MANDATORY: A pull-up resistor (4.7k to 10k) must be connected from the output pin to VCC.

The LM393 is used inside nearly every sensor module that produces a digital output threshold: soil moisture sensors, IR obstacle sensors, sound detection modules, vibration sensors. The onboard potentiometer sets the threshold, and the LM393 comparator produces the digital HIGH/LOW output.

Wiring: VREF (threshold) to IN+, signal to IN-. Pull-up 10k from OUT to VCC. When signal > VREF: output transistor OFF, pulled HIGH by resistor. When signal < VREF: output transistor ON, pulled LOW. This is the active-LOW convention used by most Arduino-compatible sensor modules.

6. LM386 — Low-Voltage Audio Power Amplifier

Package: 8-pin DIP. Supply: 4V to 12V. Output Power: 250mW at 8-ohm speaker with 6V supply, 700mW at 8-ohm with 12V. Voltage Gain: Fixed at 20 (26dB) by default. Adjustable to 200 (46dB) by placing 10uF capacitor between Pins 1 and 8. Quiescent Current: 4mA. Total Harmonic Distortion: 0.2% typical.

Pinout: Pin 1 = Gain, Pin 2 = IN- (connect 10k volume potentiometer between audio source and this pin), Pin 3 = GND, Pin 4 = GND, Pin 5 = VOUT (connect to speaker through 250uF electrolytic capacitor, positive leg to speaker), Pin 6 = VCC (4-12V), Pin 7 = Bypass (connect 10uF capacitor to GND for noise filtering), Pin 8 = Gain. The 250uF output coupling cap blocks DC from reaching the speaker.

7. NE5532 — High-Performance Dual Op-Amp (Audio Grade)

Package: 8-pin DIP. The industry-standard audio-grade dual op-amp used in professional mixing consoles, preamplifiers, and hi-fi equipment. Supply: +/-3V to +/-22V. Gain-Bandwidth: 10 MHz. Slew Rate: 9 V/us. Equivalent Input Noise Voltage: 5 nV/rtHz (extremely low noise). Output can drive 600-ohm loads directly.

Pinout (identical to LM358): Pin 1=OUT-A, Pin 2=IN-A, Pin 3=IN+A, Pin 4=V-, Pin 5=IN+B, Pin 6=IN-B, Pin 7=OUT-B, Pin 8=V+. Used in phono preamplifiers (RIAA equalization), microphone preamps, active crossover networks, and audio filters where noise is critical.